

Home work 12

Some notations: Assume ω is a string on an alphabet Σ and $a \in \Sigma$, then by $\#_a(\omega)$ we denote the number of symbols a in the string ω . For example $\#_0(01001101) = 4$.

Task 1. Construct NFA and DFA accepting the following languages:

- (a) $L = \{\omega \in \{0,1\}^* : \omega \text{ contains a substrings } 01 \text{ and } 10\}$
- (b) $L = \{\omega \in \{0,1\}^* : \omega \text{ contains a substring } 01 \text{ or } 10\}$
- (c) $L = \{\omega \in \{0,1\}^* : \omega \text{ is ended with } 01 \text{ or } 10\}$
- (d) $L = \{\omega \in \{0,1\}^* : \omega \text{ beginning with } 01 \text{ and ended with } 10\}$
- (e) $L = \{\omega \in \{0,1\}^* : \omega \text{ contains a substring } 01101\}$
- (f) $L = \{\omega \in \{0,1\}^* : \omega \text{ begins with } 011 \text{ or contains a substring } 100\}$
- (g) $L = \{\omega \in \{0,1\}^* : \omega \text{ begins with } 011 \text{ and contains a substring } 100\}$
- (h) $L = \{\omega \in \{0,1\}^* : \text{the first and last symbols of } \omega \text{ are same}\}$
- (i) $L = \{\omega \in \{0,1\}^* : \text{the first and last symbols of } \omega \text{ are different}\}$
- (j) $L = \{\omega \in \{0,1\}^* : \#_0(\omega) = 1 \pmod{3}\}$
- (k) $L = \{\omega \in \{0,1\}^* : \#_0(\omega) \text{ is even and } \#_1(\omega) \text{ is odd}\}$
- (l) $L = \{\omega \in \{0,1\}^* : \omega \text{ beginning with } 011 \text{ and } \#_0(\omega) \text{ is even}\}$
- (m) $L = \{\omega \in \{0,1\}^* : \omega \text{ beginning with } 011 \text{ or contains substring } 0101 \text{ or ended with } 010\}$
- (n) $L = \{\omega \in \{0,1\}^* : \omega \text{ beginning with } 011 \text{ and contains substring } 0101 \text{ and ended with } 010\}$
- (o) $L = \{\omega \in \{0,1\}^* : \text{number occurrences of substring } 010 \text{ in } \omega \text{ is even}\}$
- (p) $L = \{\omega \in \{0,1\}^* : \omega \text{ ended with } 011 \text{ and } \#_1(\omega) \text{ is even}\}$
- (q) $L = \{\omega \in \{0,1\}^* : \omega \text{ contain a substring } 0110 \text{ or } \#_0(\omega) \text{ is odd}\}$
- (r) $L = \{\omega \in \{0,1\}^* : \#_0(\omega) \neq \#_1(\omega) \pmod{2}\}$
- (s) $L = \{\omega \in \{0,1\}^* : \omega \text{ is binary expansion of positive integer which are congruent to zero modulo } 3\}$

- (t) $L = \{\omega \in \{0, 1\}^* : \omega \text{ is binary expansion of positive integer which are congruent to zero modulo } 6\}$
- (u) $L = \{\omega \in \{0, 1\}^* : \omega \text{ is binary expansion of positive integer which are congruent to zero modulo } 7\}$
- (v) $L = \{\omega \in \{0, 1\}^* : \omega \text{ contains a substring } 010 \text{ and does not end with } 110\}$
- (w) $L = \{\omega \in \{0, 1\}^* : \omega \text{ does not contain a substrings } 011 \text{ and } 110\}$
- (x) $L = \{\omega \in \{0, 1\}^* : \omega \text{ is ternary expansion of positive integer which are congruent to zero modulo } 5\}$
- (y) $L = \{\omega \in \{0, 1\}^* : \text{every block every block of four consecutive symbols contains a substring } 01\}$
- (z) $L = \{\omega \in \{0, 1\}^* : \text{every block every block of three consecutive symbols contains a substring } 01\}$